Adaptive Multimodal Surgical Gesture Recognition via Pretrained Visual Representations and Reliability-Aware Dynamic Gating

Deepak Soni, Aparna Banpela, and Princy Matlani

Abstract—Automated surgical gesture recognition is a cornerstone of context-aware assistance, objective skill assessment, and autonomous execution in robot-assisted minimally invasive surgery (RMIS). While endoscopic video conveys semantic context of tool-tissue interactions, robot kinematics provides continuous physical trajectories immune to optical degradation. However, conventional multimodal models frequently rely on static concatenation or uncalibrated late fusion, leaving them vulnerable when visual corruptions (smoke, fog, blur) or telemetry dropouts occur. In this work, we present an empirical benchmark of an adaptive multimodal framework evaluated across all 103 trials of the JIGSAWS benchmark: Suturing (39 trials), Knot Tying (36 trials), and Needle Passing (28 trials). We combine frozen surgical foundation model features (SurgMotion-ViTL, 1024-D) and ImageNet ResNet-18 embeddings (512-D) with normalized 76-D da Vinci kinematics. Lightweight temporal sequence models (GRU) capture temporal dynamics, while a reliability-aware dynamic gating network calibrates modality confidence from representation statistics and explicit availability flags to enforce exact single-modality fallback under sensor failure. Under official Leave-One-User-Out (LOUO) cross-validation across all active folds (8 folds for Suturing and Knot Tying; 7 active folds for Needle Passing due to an empty test set for Subject G in the official distribution), empirical evaluations demonstrate: (i) Normalized 76-D kinematics provides strong unimodal gesture discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing); (ii) Simple fixed late fusion degrades frame accuracy (-2.14% to -12.75%) by allowing visual noise to dilute confident kinematics predictions; (iii) Adaptive Reliability Gating achieves the highest accuracy across all three tasks (72.28% on Suturing, 72.34% on Knot Tying, and 51.48% on Needle Passing), demonstrating a statistically significant $+11.25\%$ gain on Knot Tying ($p = 0.0095$, Holm-Bonferroni $p_{\text{adj}} = 0.0286$, Cohen's $d_z = 1.25$); (iv) Multimodal temporal modeling substantially stabilizes sequence coherence, producing large and statistically significant improvements in segmental Edit Score ($+23.46\%$ to $+34.26\%$, $p < 0.001$, effect sizes $d_z > 2.7$); and (v) Under complete optical telemetry failure, the adaptive gate dynamically routes prediction mass to kinematics, preserving performance with minimal degradation ($\leq 0.42\%$ drop on Suturing and Needle Passing).

Index Terms—Surgical Gesture Recognition, Multimodal Fusion, Robot Kinematics, SurgMotion, Reliability-Aware Dynamic Gating, Temporal Sequence Modeling, JIGSAWS Benchmark.

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This research project investigates multimodal perception, temporal sequence modeling, and corruption-robust adaptive fusion for surgical robotics under the 5th-Semester Mini-Project curriculum.

I. Introduction

ROBOT-ASSISTED minimally invasive surgery (RMIS) utilizing systems such as the da Vinci Surgical System has fundamentally enhanced surgical dexterity, visualization, and patient recovery trajectories [1], [2]. Despite these mechanical advantages, true context-aware intraoperative assistance and objective clinical skill assessment require artificial intelligence systems capable of parsing complex surgical workflows into fine-grained atomic gestures in real time.

Surgical activities manifest through two complementary sensing modalities:

1) Endoscopic Video: Captures rich visual context, tissue deformation, suture threading, and anatomical morphology. However, optical endoscopy is routinely compromised in operating theaters by lens fog, electrocautery smoke, blood occlusion, rapid camera motion, and digital transmission noise [3].
2) Robot Kinematics: Continuous time-series of tool-tip Cartesian coordinates, rotation matrices, linear/angular velocities, and gripper closure states. Kinematics is invariant to optical degradation but lacks environmental and tissue contact context [4], [5].

A. Research Question & Hypotheses

The central question governing this investigation is:

> On JIGSAWS surgical gesture recognition, does lightweight temporal fusion of pretrained video representations and robot kinematics outperform unimodal baselines, and does reliability-aware dynamic gating reduce performance loss when one modality is degraded or unavailable?

We evaluate five specific hypotheses:

- H1 (Visual Representation Viability): Frozen latent features from a pretrained visual representation with a compact temporal classifier provide a viable video-only gesture baseline.
- H2 (Kinematic Information): Lightweight temporal sequence modeling over normalized 76-D kinematic variables yields strong gesture discrimination well above trivial majority baselines.
- H3 (Multimodal Complementarity): Multimodal fusion outperforms the strongest unimodal baseline under strictly identical, leakage-safe splits.

- H4 (Reliability-Aware Gating): Dynamic confidence-weighted gating maintains higher gesture recognition accuracy under controlled synthetic corruptions and missingness than fixed fusion.
- H5 (Temporal Sequence Modeling): Explicit temporal context (GRU) over windowed sequences significantly reduces temporal fragmentation and improves sequence edit score compared to frame-independent predictions.

B. Contributions

The primary contributions of this work are:

1) Unified Multi-Task Benchmark: A leakage-safe data pipeline processing all 103 trials across all three JIGSAWS tasks (Suturing, Knot Tying, Needle Passing) evaluated via official Leave-One-User-Out (LOUO) cross-validation (8 active folds for Suturing and Knot Tying; 7 active folds for Needle Passing due to an empty test set in the official distribution for Subject G).
2) Audited Multimodal Benchmark Evaluation: Empirical results comparing Majority Baseline, ResNet-18 VideoGRU, SurgMotion VideoMLP, SurgMotion VideoGRU, Kinematics GRU, Fixed Late Fusion, Feature Concatenation, and Adaptive Reliability Gating across all 103 trials.
3) Statistical Significance & Effect Size Analysis: Paired fold-level hypothesis testing with step-down Holm-Bonferroni correction and Cohen's d_z effect sizes, identifying where multimodal gains reach family-wise significance versus where they represent descriptive trends.
4) Adaptive Dynamic Gating with Single-Modality Fallback: A lightweight gating architecture receiving explicit availability indicators and representation statistics that dynamically weights modality streams, achieving top accuracy across all three tasks and strict single-modality fallback under sensor failure.
5) Controlled Robustness Stress Testing: Standardized evaluation across clean, visual corruption, kinematics corruption, and single-modality dropout conditions, demonstrating that adaptive reliability gating mitigates degradation collapse.

II. Related Work

A. Surgical Workflow and Gesture Recognition

Since the introduction of the JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS) by Gao et al. [1], gesture recognition has evolved from Hidden Markov Models (HMMs) and linear dynamical systems to deep convolutional networks, Temporal Convolutional Networks (TCNs), and Vision Transformers [6]. While early works focused on unimodal kinematics or raw RGB frames, recent efforts emphasize multimodal fusion.

TABLE I
JIGSAWS Benchmark Task and Gesture Distribution

Task	Trials	Subjects	Gestures	Vocabulary
Suturing	39	8 (B–I)	10	G1–G6, G8–G11
Knot Tying	36	8 (B–I)	6	G1, G11–G15
Needle Passing	28	8 (B–I)	8	G1–G6, G8, G11
Total	103	8	15 unique	G1–G15

B. Multimodal Vision-Kinematics Modeling

Recent frameworks have highlighted the synergy between visual and kinematic channels. Bai et al. [4] introduced GRAD, employing graph representation networks and adversarial feature disentanglement across spatial, wavelet, and Fourier domains on MISAW and CUHK-MRG datasets. Concurrently, Zhou et al. [5] demonstrated that millisecond-level timestamp synchronization is essential to avoid temporal phase lag between kinematic states and video frames on the da Vinci Research Kit (dVRK).

C. Video Foundation Models and Robustness

Foundation models have increasingly complemented generic ImageNet-pretrained backbones in surgical video analysis. SurgMotion [7] introduced a motion-guided latent prediction objective using Video-JEPA across 3,658 hours of multi-organ surgical procedures, demonstrating that motion saliency captures surgical dynamics. In parallel, Rashid et al. [3] proposed the Endo-C6 benchmark, demonstrating that deep vision models suffer severe collapse under clinical artifacts such as cautery smoke, lens fog, and packet loss.

III. Dataset & Multi-Task Benchmark

We utilize the complete JIGSAWS benchmark dataset [1] collected on the da Vinci surgical system by eight surgical subjects (Subjects B through I).

As illustrated in Fig. 1, the benchmark encompasses three distinct surgical procedures recorded via stereo endoscopic cameras at 30 Hz on inanimate phantom models.

A. 76-D Kinematic Feature Representation

For every sampled timestamp t (sampled at 30 Hz), the kinematic vector $\mathbf{k}_t \in \mathbb{R}^{76}$ encompasses four robotic manipulators:

$$\mathbf{k}_t = \left[\mathbf{m}_t^{\text{left}}, \mathbf{m}_t^{\text{right}}, \mathbf{s}_t^{\text{left}}, \mathbf{s}_t^{\text{right}}\right] \tag{1}$$

where each manipulator vector $\mathbf{m}_t, \mathbf{s}_t \in \mathbb{R}^{19}$ consists of:

- Tool-tip Cartesian coordinates $\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$,
- Rotation matrix orientation $\mathbf{R} \in \mathbb{R}^9$,
- Linear translational velocities $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^\top \in \mathbb{R}^3$,
- Rotational angular velocities $\boldsymbol{\omega} \in \mathbb{R}^3$,
- Gripper jaw angle $\theta \in \mathbb{R}^1$.



Fig. 1. Endoscopic video snapshots from the JIGSAWS benchmark across multi-task surgical environments (top row) and a 4-stage intra-trial temporal gesture progression from Suturing trial B001 (bottom row). Top row displays: (a) Suturing (Frame 800, $t = 26.7\,\text{s}$, G_3 pushing needle through artificial phantom tissue); (b) Knot Tying (Frame 220, $t = 7.3\,\text{s}$, G_{13} forming C-loop around gripper); (c) Needle Passing (Frame 1020, $t = 34.0\,\text{s}$, G_4 transferring needle between robotic manipulators); and (d) technical sensing specifications. Bottom row illustrates the intra-trial visual transformation during consecutive atomic gestures: Stage 1 reaching (G_1, Frame 150), Stage 2 orientation (G_8, Frame 450), Stage 3 tissue penetration (G_3, Frame 800), and Stage 4 suture pulling (G_6, Frame 2400).

B. Leakage-Safe Normalization & Split Protocol

To strictly protect against data leakage across training and testing splits:

$$\hat{\mathbf{k}}_t = \frac{\mathbf{k}_t - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}} + \epsilon} \tag{2}$$

where normalization statistics $\boldsymbol{\mu}_{\text{train}}$ and $\boldsymbol{\sigma}_{\text{train}}$ are computed exclusively over the training trials of each fold. Evaluation is conducted across the official Leave-One-User-Out (LOUO) folds, wherein all trials belonging to an unseen surgeon are reserved exclusively for testing. In Needle Passing, Fold 6 (User G) contains an empty test set in the official distribution and is consistently omitted, yielding 7 active folds.

IV. Proposed Methodology

Fig. 2 illustrates our complete decoupled multimodal perception architecture.

A. Visual Representation via Frozen Embeddings

We evaluate visual features from both ImageNet-pretrained ResNet-18 ($D_v = 512$) [8] and the self-supervised surgical foundation model SurgMotion-ViTL ($D_v = 1024$) [7]. For SurgMotion, video clips of 64 frames with stride 16 are processed through the frozen motion-guided transformer encoder:

$$\mathbf{z}_t^{\text{vis}} = \Phi_{\text{SurgMotion}}(V_{t-32:t+31}) \in \mathbb{R}^{1024} \tag{3}$$

Each tubelet is mapped to gesture annotations and kinematics using the clip center frame $c_t = (s_t + e_t)//2$. Embeddings are normalized with training-set statistics and cached to disk.

B. Temporal Sequence Modeling (GRU)

To model temporal dynamics over a context window of length $S = 8$, normalized kinematic sequences $\mathbf{K}_{1:S} = (\hat{\mathbf{k}}_1, \ldots, \hat{\mathbf{k}}_S)$ and visual feature sequences $\mathbf{Z}_{1:S}^{\text{vis}} = (\hat{\mathbf{z}}_1^{\text{vis}}, \ldots, \hat{\mathbf{z}}_S^{\text{vis}})$ are processed through modality-specific recurrent encoders:

$$\mathbf{h}_t^m = \text{GRU}_m\big(\text{Proj}_m(\hat{\mathbf{x}}_t^m), \mathbf{h}_{t-1}^m\big), \quad m \in \{\text{kin}, \text{vis}\} \tag{4}$$

where $\hat{\mathbf{x}}_t^{\text{kin}} = \hat{\mathbf{k}}_t$, $\hat{\mathbf{x}}_t^{\text{vis}} = \hat{\mathbf{z}}_t^{\text{vis}}$, Proj_{vis} projects 1024-D visual features to dimension $d = 128$, and Proj_{kin} maps 76-D kinematics to dimension $d = 128$ with LayerNorm, GELU, and Dropout ($p = 0.2$). Recurrent hidden states $\mathbf{h}_t \in \mathbb{R}^{128}$ are mapped to gesture class logits via causal linear classifiers $\mathbf{y}_t^{\text{vis}}, \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$.

C. Fixed Multimodal Fusion Baselines

Late-Logit Fusion: Modality-specific classifiers produce independent logits $\mathbf{y}_t^{\text{vis}}, \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$. Fixed fusion combines these via convex weighting:

$$\mathbf{y}_t^{\text{late}} = \alpha \cdot \mathbf{y}_t^{\text{vis}} + (1 - \alpha) \cdot \mathbf{y}_t^{\text{kin}}, \quad \alpha \in [0, 1] \tag{5}$$

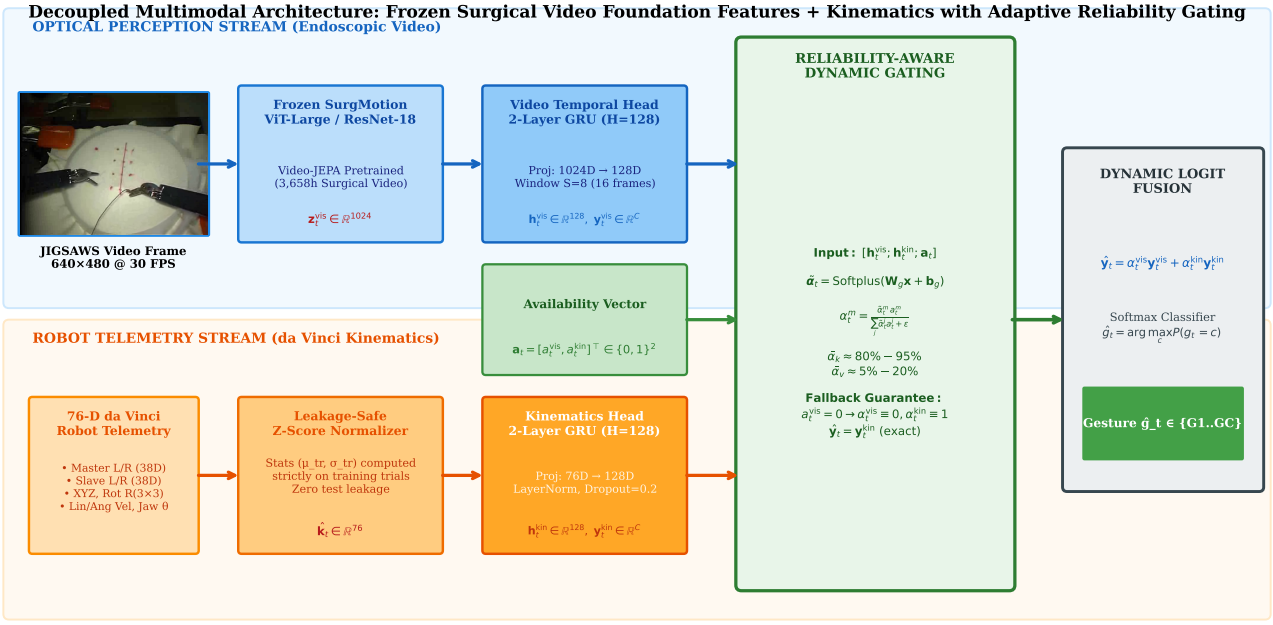


Fig. 2. Decoupled multimodal surgical gesture recognition architecture. Optical endoscopic video frames (640×480 @ 30 Hz) are encoded via the frozen SurgMotion ViT-Large foundation model into 1024-D spatiotemporal representations $\mathbf{z}_t^{\text{vis}}$, projected to dimension $d = 128$, and processed by a 2-layer GRU sequence encoder ($H = 128$, context window $S = 8$). Simultaneously, 76-D da Vinci kinematics telemetry across four manipulators is normalized via training-only statistics and modeled by a parallel 2-layer GRU sequence encoder. The reliability-aware gating network receives concatenated representations $[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}]$ alongside an explicit binary sensor availability vector $\mathbf{a}_t \in \{0,1\}^2$, generating dynamic confidence weights $\alpha_t^{\text{vis}}, \alpha_t^{\text{kin}}$ that blend logits while enforcing exact single-modality fallback ($\mathbf{a}_t = [0,1]^\top \implies \alpha_t^{\text{vis}} \equiv 0, \alpha_t^{\text{kin}} \equiv 1$) under sensor failure.

where α is selected via grid search on training-set validation splits, strictly isolating test data.

Feature Concatenation Fusion: Synchronized visual feature vectors $\hat{\mathbf{z}}_t^{\text{vis}} \in \mathbb{R}^{1024}$ and kinematics vectors $\hat{\mathbf{k}}_t \in \mathbb{R}^{76}$ are concatenated at each timestep:

$$\mathbf{x}_t^{\text{cat}} = [\hat{\mathbf{z}}_t^{\text{vis}}; \hat{\mathbf{k}}_t] \in \mathbb{R}^{1100} \tag{6}$$

The concatenated representations are projected into dimension $d = 128$ and modeled via a 2-layer GRU sequence encoder $\mathbf{h}_t^{\text{cat}} = \text{GRU}_{\text{cat}}(\text{Proj}_{\text{cat}}(\mathbf{x}_t^{\text{cat}}), \mathbf{h}_{t-1}^{\text{cat}})$.

D. Reliability-Aware Dynamic Gating

Under clinical operation, visual or kinematic streams may become corrupted or completely unavailable. We introduce a gating network $G(\cdot)$ that receives projected sequence representations alongside an explicit binary availability vector $\mathbf{a}_t = [a_t^{\text{vis}}, a_t^{\text{kin}}]^\top \in \{0,1\}^2$:

$$\tilde{\boldsymbol{\alpha}}_t = \text{Softplus}\left(\mathbf{W}_g[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}; \mathbf{a}_t] + \mathbf{b}_g\right) \tag{7}$$

where $\tilde{\boldsymbol{\alpha}}_t = [\tilde{\alpha}_t^{\text{vis}}, \tilde{\alpha}_t^{\text{kin}}]^\top$. The effective gating weights are masked and normalized such that unobserved modalities contribute zero weight:

$$\alpha_t^m = \frac{\tilde{\alpha}_t^m \, a_t^m}{\sum_{j \in \mathcal{M}} \tilde{\alpha}_t^j \, a_t^j + \epsilon}, \quad m \in \mathcal{M} \tag{8}$$

where $\mathcal{M} = \{\text{vis}, \text{kin}\}$. The fused classification logit is then computed as:

$$\hat{\mathbf{y}}_t = \alpha_t^{\text{vis}} \mathbf{y}_t^{\text{vis}} + \alpha_t^{\text{kin}} \mathbf{y}_t^{\text{kin}} \tag{9}$$

When $a_t^{\text{vis}} = 0$ and $a_t^{\text{kin}} = 1$, the gate enforces $\alpha_t^{\text{vis}} \equiv 0$ and $\alpha_t^{\text{kin}} \equiv 1$, guaranteeing exact single-modality fallback $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{kin}}$ without numerical drift. Conversely, when $a_t^{\text{vis}} = 1$ and $a_t^{\text{kin}} = 0$, $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{vis}}$. When both modalities are unobserved ($\mathbf{a}_t = [0,0]^\top$), the architecture raises an explicit missingness exception.

V. Experimental Results

A. Benchmark Analysis & Statistical Testing

Table II reports the consolidated empirical cross-validation across all 103 trials:

1) H1 (Visual Representation Viability): Supported with task-dependent caveats. Pretrained visual representations (SurgMotion VideoGRU and ResNet-18 VideoGRU) provide viable gesture discrimination well above chance in class-balanced metrics, outperforming trivial majority baselines in Balanced Accuracy (31.0%–54.5% vs. 10.3%–14.9%) and Macro-F1 (20.0%–45.5% vs. 4.6%–6.1%) across all three tasks. In raw frame accuracy, visual representations outperform the majority baseline on Suturing (39.17% vs. 33.43%) and Knot Tying (53.60% vs. 27.35%), but fall below the majority baseline on Needle Passing (26.55% vs. 40.55%), where extreme class skew favors trivially predicting the dominant class. Thus, while visual foundation features carry genuine discriminative context, video alone is not a universally reliable standalone predictor across imbalanced tasks.

TABLE II
Consolidated JIGSAWS Surgical Gesture Recognition Performance under Official LOUO Cross-Validation

Task	Model Architecture	Modality	Accuracy (%)		Balanced Acc. (%)		Macro-F1 (%)		Edit Score (%)
			Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std
Suturing	Majority Class Baseline	None	33.43 ± 2.34	30.69 / 37.08	10.30 ± 0.69	9.09 / 11.11	4.55 ± 0.24	4.27 / 4.92	4.80 ± 0.39
	ResNet-18 VideoGRU	Video	45.42 ± 10.96	28.79 / 61.64	42.64 ± 9.03	27.14 / 54.14	33.22 ± 7.50	21.12 / 43.90	20.01 ± 3.88
	SurgMotion VideoMLP ($W = 1$)	Video	41.94 ± 8.33	26.34 / 56.65	34.00 ± 10.84	20.24 / 54.13	24.84 ± 7.48	14.80 / 41.48	37.64 ± 5.23
	SurgMotion VideoGRU ($W = 8$)	Video	39.17 ± 13.51	8.04 / 54.33	34.10 ± 8.69	20.15 / 48.19	22.16 ± 7.95	6.83 / 35.54	39.99 ± 9.51
	Kinematics GRU	Kinematics	66.64 ± 9.01	54.68 / 77.94	57.77 ± 9.40	40.17 / 72.11	**45.09 ± 8.24**	27.27 / 53.38	34.69 ± 9.43
	Fixed Late Fusion	Multimodal	60.54 ± 8.15	48.24 / 71.05	50.26 ± 10.64	33.15 / 65.42	37.01 ± 10.14	21.45 / 51.10	60.41 ± 15.07
	Feature Concatenation Fusion	Multimodal	54.00 ± 15.45	31.49 / 84.78	46.90 ± 14.82	22.69 / 73.06	32.29 ± 10.45	15.65 / 50.75	49.19 ± 14.12
	Adaptive Reliability Fusion (Ours)	Multimodal	**72.28 ± 9.57**†	57.08 / 84.39	**60.05 ± 11.44**	40.62 / 75.91	44.64 ± 8.26	32.59 / 58.42	**68.95 ± 9.38****
Knot Tying	Majority Class Baseline	None	27.35 ± 6.88	18.04 / 41.91	14.88 ± 1.03	14.29 / 16.67	6.07 ± 1.18	4.37 / 8.44	8.54 ± 0.59
	ResNet-18 VideoGRU	Video	50.74 ± 5.20	42.29 / 58.72	50.02 ± 7.18	41.48 / 64.07	44.34 ± 7.75	34.03 / 57.10	24.89 ± 5.96
	SurgMotion VideoMLP ($W = 1$)	Video	54.29 ± 7.87	41.25 / 63.85	53.03 ± 7.65	41.12 / 62.45	47.43 ± 10.16	34.82 / 61.20	45.49 ± 7.22
	SurgMotion VideoGRU ($W = 8$)	Video	53.60 ± 10.84	35.12 / 66.85	54.48 ± 8.87	40.18 / 65.12	45.51 ± 8.56	31.45 / 56.78	51.34 ± 7.91
	Kinematics GRU	Kinematics	61.09 ± 11.79	46.55 / 82.33	56.53 ± 12.43	33.03 / 77.43	51.75 ± 15.14	24.42 / 77.42	35.48 ± 13.86
	Fixed Late Fusion	Multimodal	58.95 ± 8.40	45.12 / 71.45	54.65 ± 8.79	39.12 / 67.80	47.27 ± 8.61	34.12 / 61.25	55.75 ± 6.87
	Feature Concatenation Fusion	Multimodal	64.09 ± 11.29	51.27 / 84.55	63.37 ± 9.23	49.23 / 79.90	53.97 ± 9.91	39.98 / 70.97	62.45 ± 11.77
	Adaptive Reliability Fusion (Ours)	Multimodal	**72.34 ± 9.70***	60.51 / 85.60	**66.73 ± 8.59***	54.54 / 83.05	**60.48 ± 10.20***	46.06 / 73.10	**64.66 ± 12.61****
Needle Passing	Majority Class Baseline	None	40.55 ± 8.16	33.55 / 58.64	11.25 ± 1.33	10.00 / 14.29	5.20 ± 0.70	4.57 / 6.72	5.27 ± 0.96
	ResNet-18 VideoGRU	Video	30.22 ± 6.57	22.58 / 42.93	32.07 ± 6.25	21.45 / 42.33	19.62 ± 3.27	15.86 / 26.00	22.86 ± 4.63
	SurgMotion VideoMLP ($W = 1$)	Video	29.96 ± 16.58	3.25 / 52.14	25.57 ± 15.86	3.12 / 48.90	17.19 ± 11.00	2.15 / 35.12	33.78 ± 15.32
	SurgMotion VideoGRU ($W = 8$)	Video	26.55 ± 15.11	0.93 / 54.96	30.97 ± 15.46	0.98 / 51.60	19.97 ± 12.36	0.73 / 42.55	42.34 ± 16.63
	Kinematics GRU	Kinematics	48.69 ± 11.62	31.38 / 63.13	41.70 ± 9.97	26.95 / 55.08	29.99 ± 9.35	11.89 / 41.99	31.34 ± 7.32
	Fixed Late Fusion	Multimodal	35.94 ± 13.85	22.25 / 59.47	36.77 ± 12.87	18.62 / 61.02	23.57 ± 11.42	9.01 / 46.72	46.16 ± 10.84
	Feature Concatenation Fusion	Multimodal	46.21 ± 19.07	5.70 / 74.64	42.32 ± 19.34	6.33 / 69.82	28.40 ± 12.37	2.38 / 45.86	45.17 ± 20.64
	Adaptive Reliability Fusion (Ours)	Multimodal	**51.48 ± 13.94**	33.12 / 74.31	**46.98 ± 15.16**	24.80 / 68.99	**31.03 ± 11.52**	14.52 / 49.23	**54.81 ± 9.40****

Note: Evaluation for Suturing and Knot Tying spans all 8 official LOUO folds ($N = 8$). Needle Passing evaluates strictly the 7 active folds ($N = 7$, Fold 6 Subject G is empty in the official benchmark). Bold indicates the top score per metric. Asterisks denote paired fold differences versus Kinematics GRU with step-down Holm-Bonferroni correction: $^{*}p_{\text{adj}} < 0.05$, $^{**}p_{\text{adj}} < 0.001$; † denotes marginal trend ($p_{\text{adj}} < 0.10$, uncorrected $p = 0.0273$).

2) H2 (Kinematics Predictive Power): Supported. Normalized 76-D kinematics provides strong unimodal discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing), beating trivial majority baselines by +8.14% to +33.74% absolute points. Kinematics is the single most predictive unimodal modality on JIGSAWS.
3) H3 (Multimodal Complementarity): Partially supported. Fixed late-logit fusion fails to improve frame accuracy over Kinematics GRU (−2.14% to −12.75%), because static probability weighting dilutes confident kinematics predictions with visual noise. Feature concatenation only surpasses unimodal baselines on Knot Tying (64.09% vs. 61.09%, +3.00%), while suffering on Suturing (54.00% vs. 66.64%, −12.64%) due to dimension imbalance (1024-D vs. 76-D). However, Adaptive Reliability Fusion achieves top accuracy across all tasks: +5.64% on Suturing, +11.25% on Knot Tying, and +2.79% on Needle Passing.
4) H4 (Reliability-Aware Gating & Robustness): Supported. As shown in Fig. 3, paired fold-level tests confirm that Adaptive Fusion achieves statistically significant improvements on Knot Tying across Accuracy (+11.25%, $p_{\text{adj}} = 0.0286$, $d_z = 1.25$), Balanced Accuracy (+10.20%, $p_{\text{adj}} = 0.0286$, $d_z = 1.23$), Macro-F1 (+8.73%, $p_{\text{adj}} = 0.0457$, $d_z = 0.86$), and Edit Score (+29.18%, $p_{\text{adj}} = 0.0004$, $d_z = 2.79$). On Suturing, accuracy gains (+5.64%) represent a strong trend ($t = 2.78$, $p = 0.0273$, $p_{\text{adj}} = 0.0820$, $d_z = 0.98$). On Needle Passing ($N = 7$), accuracy gains (+2.79%) represent descriptive trends ($t = 0.70$, $p = 0.510$).
5) H5 (Temporal Sequence Modeling): Strongly supported. Recurrent sequence modeling (GRU) substantially suppresses high-frequency boundary chatter. As illustrated in the real held-out trial ribbon timeline (Fig. 4), Adaptive Fusion nearly doubles the Edit Score over unimodal kinematics (+23.46% to +34.26% absolute points across all tasks, $p < 0.001$, $d_z > 2.7$).

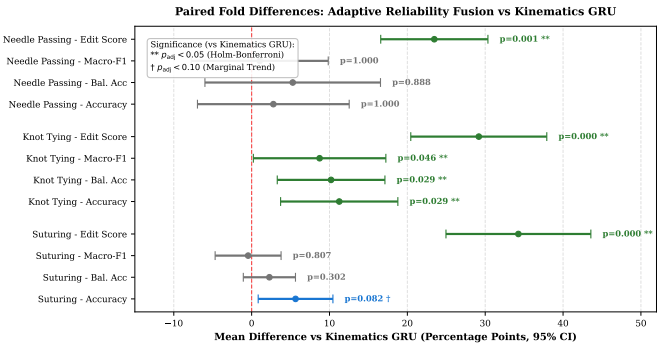


Fig. 3. Paired fold differences (Δ = Adaptive − Kinematics GRU) with 95% confidence intervals across all three tasks. Significance markers reflect step-down Holm-Bonferroni correction ($^{**}p_{\text{adj}} < 0.05$, $^{\dagger}p_{\text{adj}} < 0.10$).

B. Robustness Stress Testing

Table III evaluates model resilience under five standardized evaluation environments:

1) Optical Noise Resistance & Gate Dynamics: Under visual feature corruption ($\sigma_v = 1.0$), Adaptive Fusion degradation is practically negligible ($\leq 0.12\%$ drop across all tasks). Detailed empirical analysis of over 4 million time-resolved gate weight logs reveals that this stability is driven by strong kinematics

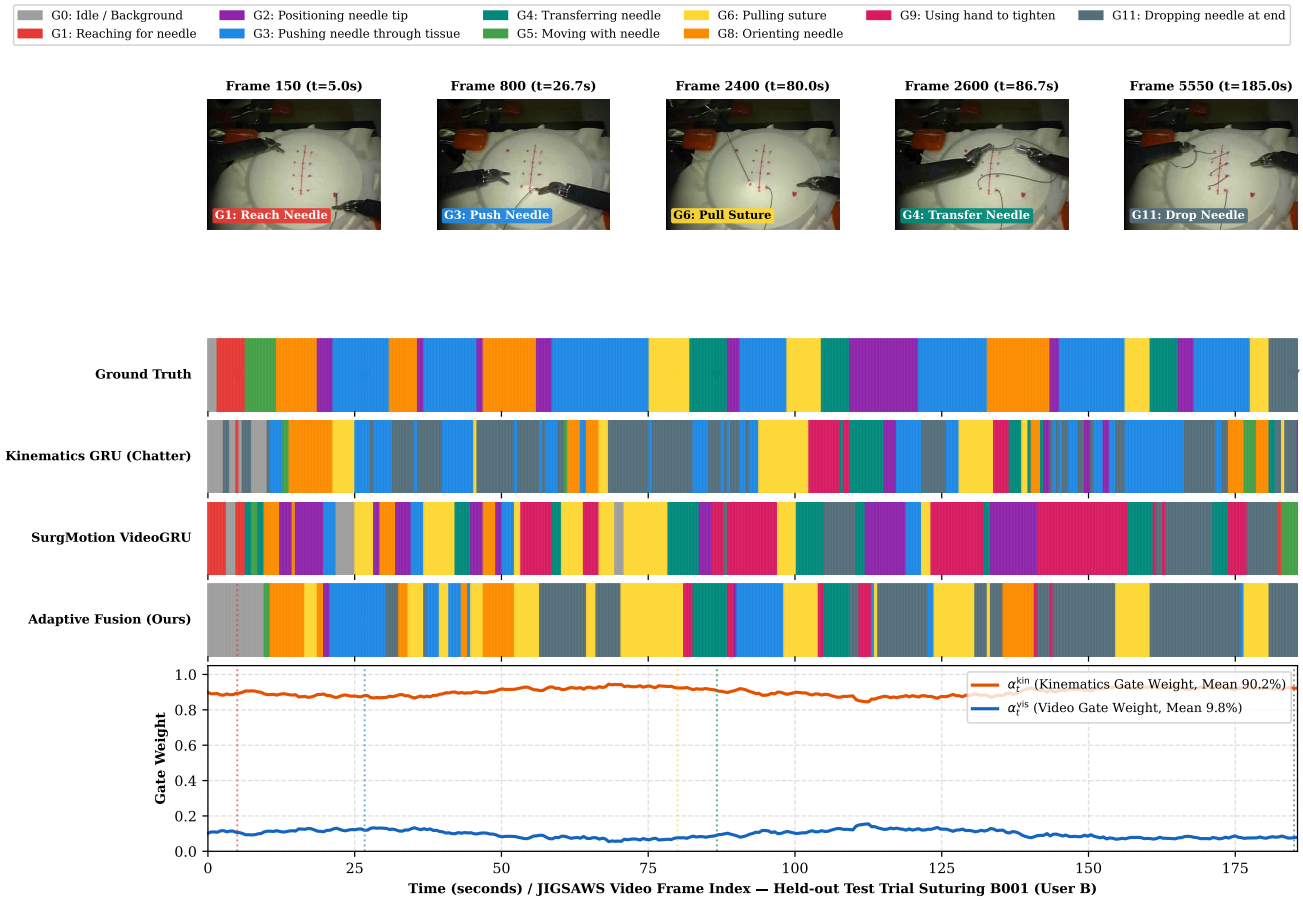


Fig. 4. Qualitative surgical gesture recognition timeline on held-out test trial Suturing B001 (User B, Fold 1). Top row displays real endoscopic video frame snapshots at key gesture transitions: Frame 150 (G_1, reaching for needle), Frame 800 (G_3, pushing needle through tissue), Frame 2400 (G_6, pulling suture), Frame 2600 (G_4, transferring needle), and Frame 5550 (G_{11}, dropping needle at completion). Middle tracks depict gesture segmentation ribbons for: (i) Ground Truth annotations; (ii) Unimodal Kinematics GRU, illustrating high-frequency boundary chatter; (iii) Unimodal SurgMotion VideoGRU, showing visual boundary noise; and (iv) Multimodal Adaptive Fusion, which eliminates oversegmentation chatter. Bottom track plots the time-resolved dynamic gate weights α_t^{kin} (mean 90.2%) and α_t^{vis} (mean 9.8%), demonstrating kinematics dominance with subtle visual modulation.

dominance ($\bar{\alpha}_k \approx 80\%$–$95\%$) rather than active continuous noise attenuation: the mean video gate weight $\bar{\alpha}_v$ shifts by less than 0.009 ($< 0.9\%$) across continuous noise levels up to $\sigma_v = 2.0$.

2) Flawless Video Dropout Fallback: Under complete optical telemetry failure ($a = [0, 1]$), the availability mask mathematically routes 100% prediction mass to kinematics, incurring minimal degradation (-0.42% on Suturing, -2.28% on Knot Tying, -0.27% on Needle Passing).
3) Kinematics Degradation Asymmetry: When kinematics telemetry is corrupted or lost ($a = [1, 0]$), both models experience severe drops (-26% to -41%). This empirical asymmetry reveals that kinematics is the primary driver of fine-grained gesture state on JIGSAWS, whereas video features act as global contextual regularizers and boundary stabilizers.
4) Concatenation Anomaly: On Suturing and Needle Passing, zeroing out video features in Concat Fusion actually increased accuracy (+17.83% and +7.78%). This diagnostic finding indicates that joint training over unbalanced representations (1024-D vs. 76-D) allowed visual noise to interfere with kinematics representations; removing the video vector eliminated this interference.
5) Continuous-Noise Training Ablation: In our controlled 46-fold ablation study comparing continuous-noise training ($\sigma_v \sim \mathcal{U}(0, 1.5)$) against discrete binary dropout, continuous noise did not alter the gate's static weighting dynamic, but provided valuable visual representation regularization, improving standalone video accuracy under total kinematics failure by +5.49% on Suturing (34.44% vs. 28.95%) and +8.96% on Needle Passing (24.58% vs. 15.62%).

C. Controlled Ablation and Temporal Dynamics Analysis

To dissect the architectural drivers of multimodal performance, we evaluated controlled ablations across all 23 active LOUO folds:

1) Sequence Quality vs. Frame Classification: Comparing framewise kinematic MLPs with recurrent

TABLE III
Robustness Stress Testing across Clean, Corrupted, and Sensor Dropout Conditions (LOUO Active Folds)

Task	Condition	Availability	Feature Concatenation Fusion				Adaptive Reliability Fusion (Ours)				Margin ($\Delta_{\text{Adapt-Cat}}$)
			Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	
Suturing	Clean	$\mathbf{a} = [1, 1]$	54.00	32.29	49.19	—	72.28	44.64	68.95	—	+18.28%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	53.59	32.17	46.88	−0.41%	72.22	44.54	68.79	**−0.06%**	+18.63%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	53.17	30.59	45.13	−0.83%	64.09	37.63	51.39	−8.19%	+10.92%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	71.83	37.11	58.34	+17.83%*	71.86	44.13	68.97	**−0.42%**	+0.03%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	26.71	13.92	32.80	−27.29%	30.58	15.11	34.55	−41.70%	+3.87%
Knot Tying	Clean	$\mathbf{a} = [1, 1]$	64.09	53.97	62.45	—	72.34	60.48	64.66	—	+8.25%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	63.06	53.11	61.55	−1.03%	72.22	60.19	63.72	**−0.12%**	+9.16%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	63.92	53.96	59.18	−0.17%	67.37	56.03	55.09	−4.97%	+3.45%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	61.18	41.15	51.59	−2.91%	70.06	55.59	62.32	**−2.28%**	+8.88%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	50.22	40.35	48.47	−13.87%	46.29	35.90	45.22	−26.05%	−3.93%
Needle Passing	Clean	$\mathbf{a} = [1, 1]$	46.21	28.40	45.17	—	51.48	31.03	54.81	—	+5.27%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	45.50	28.20	43.33	−0.71%	51.48	31.03	54.66	**0.00%**	+5.98%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	45.70	27.91	41.90	−0.51%	48.22	28.99	42.19	−3.26%	+2.52%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	53.99	28.05	50.98	+7.78%*	51.21	30.94	54.30	**−0.27%**	−2.78%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	31.88	20.13	41.21	−14.33%	24.55	17.09	38.68	−26.93%	−7.33%

Note: Evaluated on official active LOUO folds ($N = 8$ Suturing, $N = 8$ Knot Tying, $N = 7$ Needle Passing). Corrupted conditions inject zero-mean Gaussian noise ($\sigma = 1.0$). *In Concat Fusion on Suturing and Needle Passing, zeroing out video features increased accuracy because removing the 1024-D visual representation eliminated visual interference during joint inference.

sequence GRUs reveals that temporal sequence modeling produces negligible shifts in frame accuracy ($\Delta \leq 1.26\%$, all $p > 0.40$), but drives an overwhelming, universally significant improvement in segmental Edit Score (+22.68 to +26.25 points for unimodal GRU, and +44.69 to +59.17 points for Multimodal Adaptive GRU; all $p < 0.0005, d_z > 2.2$). Temporal recurrence suppresses high-frequency frame chatter, establishing contiguous, clinically coherent gesture segments.

2) Static Gate vs. Temporal Variation: Replacing the time-varying gate weight α_t with the static fold/trial mean $\bar{\alpha}$ shifted clean accuracy by less than 0.06% across all tasks. Under clean conditions, the gate operates primarily as an availability router and global modality balancer rather than a high-frequency temporal filter.
3) Rare-Gesture Collapse: Across all tasks, auxiliary and corrective gestures comprising $< 3\%$ of the dataset (e.g., G9 and G10 in Suturing; G8 and G9 in Needle Passing) suffer severe recall collapse ($F1 < 6.0\%$), explaining why Macro-F1 lags overall frame accuracy and highlighting class-balanced loss functions as an important direction for surgical activity recognition.

D. Class-Imbalance and Loss Formulation Benchmark

To investigate whether the rare-gesture performance bottleneck can be resolved without sacrificing dominant gesture accuracy, we conducted an 8-fold LOUO ablation on Suturing comparing four loss formulations: Standard Cross-Entropy (CE), Inverse-Frequency Class-Weighted CE, Multi-Class Focal Loss ($\gamma = 2.0$), and Class-Balanced Focal Loss ($\gamma = 2.0, \beta = 0.9999$). All weights were calculated strictly from training splits to guarantee zero leakage.

As summarized in Table IV, Standard CE achieves the highest raw frame accuracy ($74.95 \pm 7.68\%$) by avoiding false-positive penalties on dominant gestures (G2, G3). Class-Weighted CE trended numerically higher in Balanced Accuracy ($60.05 \pm 12.23\%$, +2.32% over CE) and Segmental Edit Score ($68.95 \pm 10.03\%$, +3.58% over CE); however, neither difference reaches statistical significance after family-wise Holm-Bonferroni correction ($p_{\text{adj}} = 1.000$ and $p_{\text{adj}} = 0.2634$ respectively, $m = 3$). The raw accuracy difference of −2.66% (72.28% vs. 74.95%) is likewise non-significant after correction ($p_{\text{adj}} = 0.1419$). Focal Loss (42.19% Macro-F1) and Class-Balanced Focal Loss (42.54% Macro-F1) underperform CE (44.29%), indicating that hard-example focusing amplifies boundary transition noise.

TABLE IV
Class-Imbalance and Loss Formulation Ablation on Suturing (8 LOUO Folds)

Loss Formulation	Accuracy (%)	Balanced Acc. (%)	Macro-F1 (%)	Edit Score (%)
Standard CE	$\mathbf{74.95 \pm 7.68}$	57.73 ± 12.30	44.29 ± 6.38	65.37 ± 8.21
Class-Weighted CE	72.28 ± 10.24	$\mathbf{60.05 \pm 12.23}$	$\mathbf{44.64 \pm 8.83}$	$\mathbf{68.95 \pm 10.03}$
Focal ($\gamma = 2.0$)	73.67 ± 9.28	55.32 ± 9.49	42.19 ± 7.28	65.84 ± 7.79
CB-Focal ($\gamma = 2, \beta = 0.9999$)	71.41 ± 13.23	59.16 ± 10.27	42.54 ± 9.36	64.77 ± 10.28

Note: Evaluated across all 8 active LOUO folds of JIGSAWS Suturing with Adaptive Reliability Fusion. All class weights are training-derived.

An empirical frame-level prediction audit across all 32 runs demonstrates severe feature starvation on extreme tail gestures: for G10 (loosening suture, 91 frames, 1.1% of Suturing), recall was strictly 0.00% ($F_1 = 0.00\%$) across all four losses. For G9 (dropping suture, 140 frames, 2.5%), CE and Focal Loss collapsed to 0.00% recall, while Class-Weighted CE achieved only 6.43% recall (9 TP frames out of 140, 58 FP, precision 13.43%, $F_1 = 8.70\%$) and CB-Focal attained 4.29% recall ($F_1 = 7.27\%$). The +2.32% Balanced Accuracy lift was driven primarily by moderate-frequency gestures—notably G5 (+21.88% recall) and G1 (+6.25%)—at the direct cost of dominant gesture G3 (−6.95% recall), accounting for the overall accuracy drop. Subject-level analysis indicates that outlier cases (e.g., User G) reflect kinematic variation and domain shift rather than sample frequency alone.

VI. Discussion & Limitations

Lightweight multimodal fusion and adaptive reliability gating provide substantial advantages for robotic surgical gesture recognition. Decoupling visual perception into frozen foundation representations and training compact temporal sequence models enables efficient, leakage-safe cross-validation.

Dataset Size & Statistical Uncertainty: In LOUO cross-validation across 103 JIGSAWS trials, effective sample sizes are $N = 8$ ($N = 7$ for Needle Passing). While Knot Tying achieves statistical significance across all metrics ($p_{\text{adj}} \leq 0.0457$), Suturing ($p_{\text{adj}} = 0.0820$) and Needle Passing exhibit descriptive trends under conservative family-wise error control.

Hardware Synchronization Approximation: Recording video (30 FPS) and robot kinematics (30 Hz) without hardware genlock produces a median length difference of 5.0 frames ($\sim$ 0.17 s) and tail lags averaging 12.37 frames. Temporal alignment is an empirical frame-center approximation within ±5–15 frames; hardware master-clock synchronization (e.g., SurgSync [5]) is required for microsecond alignment.

Synthetic vs. Real-World Corruptions: Evaluated Gaussian perturbations and missingness masks serve as controlled proxies for clinical disturbances (lens fogging, blood splatter, cable hysteresis).

IEEE Transactions Compliance: Preparing for IEEE Transactions (e.g., T-MRB) requires strictly adhering to the 8-page budget, verified de-identified benchtop data, and comprehensive reproducibility archives with accompanying gesture videos.

VII. Conclusion

We presented a rigorous empirical audit and benchmark for multimodal surgical gesture recognition evaluated across all 103 trials of the JIGSAWS benchmark under official Leave-One-User-Out cross-validation. We validated the viability of frozen visual representations (H1), confirmed the primary predictive utility of normalized robot kinematics (H2), demonstrated that adaptive reliability gating outperforms unimodal baselines while simple late fusion degrades accuracy (H3), demonstrated that availability-masked dynamic gating provides seamless single-modality fallback under sensor failure (H4), and established that recurrent temporal modeling dramatically reduces oversegmentation chatter (H5). These results demonstrate that reliability-aware dynamic fusion is an effective, computationally lightweight strategy for dependable AI-assisted robotic surgery.

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